

## ADA, OK, USA

WMO: 722044

Lat: 34.804N

Lon: 96.671W

Elev: 310

StdP: 97.66

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53930

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.9       | -6.1 | -15.2                         | 1.0 | -6.7 | -12.4 | 1.3 | -0.6 | 10.7                  | 7.8  | 9.3 | 8.4  | 2.8                   | 340  | 0.468 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 37.6            | 23.5 | 36.1 | 23.5 | 34.1 | 23.3 | 25.8                | 32.4 | 25.3 | 32.1 | 24.8 | 31.4 | 3.9                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.1                            | 19.7 | 28.7 | 23.0 | 18.4 | 27.6 | 22.8 | 18.2 | 27.5 | 81.5          | 32.9 | 79.4 | 31.7 | 76.8 | 31.8 | 28.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.4           | 8.5  | 7.8 | -13.3                      | 39.3 | 2.8                | 2.1 | -15.4                                              | 40.8 | -17.0      | 42.0 | -18.6      | 43.2 | -20.6      | 44.8 |    |    |
| (5)               |      |     | -13.6                      | 26.7 | 2.7                | 0.8 | -15.5                                              | 27.3 | -17.1      | 27.7 | -18.7      | 28.2 | -20.6      | 28.8 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 16.9  | 5.0   | 7.2   | 12.3  | 16.4  | 20.6  | 25.9  | 27.9  | 27.6  | 23.8  | 17.6  | 11.6  | 6.0  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.32  | 5.88  | 6.27  | 5.63  | 4.64  | 4.12  | 2.67  | 2.90  | 3.16  | 3.55  | 4.76  | 5.32  | 5.29 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 529   | 173   | 117   | 41    | 7     | 1     | 0     | 0     | 0     | 0     | 3     | 43    | 144  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1706  | 412   | 314   | 198   | 88    | 26    | 0     | 0     | 4     | 73    | 210   | 382   | (9)  |      |
| (10) |                                                                                  | CDD10.0     | 3038  | 19    | 38    | 112   | 199   | 329   | 476   | 555   | 547   | 415   | 237   | 89    | 22   | (10) |
| (11) |                                                                                  | CDD18.3     | 1173  | 0     | 2     | 11    | 29    | 95    | 226   | 297   | 288   | 169   | 49    | 7     | 1    | (11) |
| (12) |                                                                                  | CDH23.3     | 12482 | 4     | 16    | 71    | 237   | 777   | 2362  | 3539  | 3373  | 1619  | 424   | 58    | 2    | (12) |
| (13) |                                                                                  | CDH26.7     | 5819  | 0     | 2     | 13    | 51    | 236   | 1063  | 1868  | 1767  | 691   | 120   | 8     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.5   | 3.6   | 3.8   | 4.2   | 4.3   | 3.7   | 3.5   | 2.9   | 2.8   | 2.9   | 3.3   | 3.8   | 3.5  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1017  | 45    | 53    | 82    | 95    | 134   | 107   | 72    | 78    | 117   | 102   | 71    | 59   | (15) |
| (16) |                                                                                  | PrecMax     | 1625  | 187   | 176   | 204   | 289   | 368   | 274   | 217   | 283   | 317   | 316   | 174   | 168  | (16) |
| (17) |                                                                                  | PrecMin     | 597   | 0     | 1     | 7     | 9     | 6     | 21    | 0     | 0     | 13    | 0     | 8     | 6    | (17) |
| (18) |                                                                                  | PrecStd     | 226   | 36    | 40    | 49    | 56    | 78    | 59    | 56    | 57    | 74    | 84    | 48    | 43   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 22.9  | 26.0  | 27.9  | 31.3  | 33.7  | 36.4  | 40.1  | 40.3  | 36.1  | 32.5  | 27.7  | 22.7 | (19) |
| (20) |                                                                                  |             | MCWB  | 12.8  | 15.3  | 17.4  | 18.7  | 22.1  | 22.4  | 23.2  | 22.9  | 23.1  | 21.4  | 18.1  | 15.0 | (20) |
| (21) |                                                                                  | 2%          | DB    | 20.0  | 22.3  | 25.2  | 27.6  | 31.2  | 34.8  | 37.6  | 37.8  | 33.8  | 29.9  | 25.0  | 19.9 | (21) |
| (22) |                                                                                  |             | MCWB  | 12.3  | 14.4  | 16.8  | 18.4  | 22.2  | 23.1  | 23.5  | 23.4  | 22.9  | 21.5  | 16.2  | 13.6 | (22) |
| (23) |                                                                                  | 5%          | DB    | 17.4  | 19.2  | 22.8  | 26.2  | 29.0  | 32.8  | 36.2  | 36.2  | 32.4  | 27.6  | 22.4  | 17.2 | (23) |
| (24) |                                                                                  |             | MCWB  | 10.9  | 13.7  | 15.8  | 17.8  | 21.4  | 23.2  | 23.6  | 23.8  | 22.5  | 19.4  | 15.8  | 12.0 | (24) |
| (25) |                                                                                  | 10%         | DB    | 14.7  | 17.2  | 21.3  | 24.0  | 27.5  | 32.2  | 34.1  | 34.0  | 31.1  | 26.0  | 20.1  | 14.8 | (25) |
| (26) |                                                                                  |             | MCWB  | 9.0   | 12.1  | 15.1  | 16.8  | 20.7  | 23.2  | 23.5  | 24.0  | 22.1  | 18.6  | 14.7  | 10.6 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 16.9  | 17.9  | 19.9  | 22.0  | 24.5  | 26.6  | 26.7  | 26.9  | 25.5  | 24.0  | 20.2  | 18.4 | (27) |
| (28) |                                                                                  |             | MCDWB | 19.0  | 21.4  | 24.8  | 26.0  | 29.8  | 32.6  | 32.9  | 33.7  | 32.6  | 29.0  | 25.1  | 20.7 | (28) |
| (29) |                                                                                  | 2%          | WB    | 14.2  | 16.4  | 18.7  | 20.8  | 23.4  | 25.4  | 25.7  | 25.9  | 24.6  | 23.1  | 18.8  | 15.9 | (29) |
| (30) |                                                                                  |             | MCDWB | 17.8  | 19.6  | 22.8  | 24.7  | 28.6  | 31.3  | 32.5  | 32.7  | 30.9  | 27.1  | 23.0  | 17.8 | (30) |
| (31) |                                                                                  | 5%          | WB    | 11.7  | 14.8  | 17.6  | 19.9  | 22.6  | 24.8  | 25.2  | 25.3  | 23.9  | 21.9  | 17.1  | 13.0 | (31) |
| (32) |                                                                                  |             | MCDWB | 16.0  | 18.6  | 20.7  | 23.9  | 27.6  | 30.6  | 32.1  | 32.4  | 29.7  | 25.7  | 20.5  | 15.3 | (32) |
| (33) |                                                                                  | 10%         | WB    | 9.5   | 12.5  | 16.4  | 18.8  | 21.8  | 24.2  | 24.6  | 24.8  | 23.2  | 20.5  | 15.5  | 10.8 | (33) |
| (34) |                                                                                  |             | MCDWB | 14.0  | 16.9  | 20.2  | 22.9  | 26.3  | 29.7  | 31.5  | 31.5  | 28.5  | 23.7  | 19.1  | 13.4 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 11.1  | 11.1  | 11.5  | 11.6  | 10.2  | 10.4  | 11.1  | 11.0  | 11.6  | 12.0  | 11.8  | 10.3 | (35) |
| (36) |                                                                                  |             | MCDBR | 15.1  | 14.6  | 13.2  | 13.0  | 11.9  | 11.6  | 12.8  | 12.9  | 13.2  | 13.2  | 14.0  | 14.3 | (36) |
| (37) |                                                                                  |             | MCWBR | 9.1   | 8.7   | 7.3   | 6.9   | 5.5   | 4.1   | 3.6   | 3.6   | 5.2   | 6.4   | 7.7   | 9.5  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 13.7  | 12.2  | 10.6  | 10.8  | 10.2  | 10.1  | 11.1  | 11.1  | 11.0  | 10.5  | 12.3  | 11.7 | (38) |
| (39) |                                                                                  |             | MCWBR | 9.9   | 8.6   | 6.5   | 7.2   | 5.4   | 4.7   | 4.3   | 4.2   | 4.8   | 5.8   | 8.8   | 10.2 | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.300 | 0.307 | 0.346 | 0.376 | 0.410 | 0.423 | 0.429 | 0.434 | 0.388 | 0.345 | 0.320 | 0.300 | (40) |      |
| (41) |                                                                                  | taud        | 2.503 | 2.501 | 2.417 | 2.346 | 2.316 | 2.316 | 2.341 | 2.294 | 2.382 | 2.494 | 2.501 | 2.509 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 912   | 943   | 927   | 909   | 878   | 862   | 854   | 844   | 870   | 887   | 879   | 888   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 83    | 93    | 110   | 124   | 129   | 129   | 125   | 129   | 113   | 93    | 83    | 78    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.72  | 3.27  | 4.31  | 5.30  | 5.72  | 6.54  | 6.69  | 6.09  | 5.04  | 3.86  | 2.88  | 2.33  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.28  | 0.49  | 0.29  | 0.40  | 0.59  | 0.69  | 0.37  | 0.37  | 0.49  | 0.54  | 0.36  | 0.28  | (45) |      |

## Historical Trends

|                              |       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (47) Regional (40 neighbors) |       | (m)     |        |         |       |       |             |         |         |         |
|                              |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                              |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page